Software Requirements Specification (SRS)

SRS for Online Music Application

Problem Statement:

Listening to music using offline storage devices is inconvenient due to limited storage, lack of variety, and difficulty in discovering new songs. Similarly, artists face challenges in reaching a wide audience. Hence, there is a requirement for an online music application where users can easily access music anytime and anywhere.

Abstract:

An Online Music Application is a digital platform that enables users to stream, download, organize, and share music over the internet. The system provides a user-friendly interface where listeners can explore a vast library of songs categorized by genre, artist, album, and language. Key features typically include personalized recommendations, playlist creation, offline listening, and social sharing options. The application leverages cloud storage and advanced algorithms to ensure seamless access, high-quality audio streaming, and tailored user experiences based on listening history. Security measures are implemented to protect user data and prevent unauthorized distribution of copyrighted content. Overall, an Online Music Application enhances music accessibility, promotes artist reach, and transforms how people discover and enjoy music in the digital era.

Introduction

Online Music Application is a software system that allows users to listen to music online, search for songs, create playlists, follow artists, and receive personalized recommendations. Users can stream music, manage their accounts, and access premium features such as offline downloads. Artists can upload music content, and administrators can manage users and music data through a centralized platform.

- **Purpose**

The main purpose of the Online Music Application is to provide users with easy access to a wide collection of music with minimal effort. The application enables users to stream songs, create playlists, and get recommendations, while administrators manage content and subscriptions efficiently.

- **Definitions, Acronyms, and Abbreviations**

Streaming – Playing music online without downloading

Playlist – Collection of songs created by a user **Premium**

User – Paid user with additional features

Offline Mode – Listening to downloaded songs without internet

Recommendation Engine – System that suggests songs

- **Product Scope**

The Online Music Application provides centralized access to music and reduces dependency on offline storage. It offers:

• Time saving

• Increased efficiency

• Easy accessibility

• Organized music content

• Reduced storage usage

- **References**

- R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 8th ed. New York, NY, USA: McGraw-Hill Education, 2015.

- I. Sommerville, *Software Engineering*, 10th ed. Boston, MA, USA: Pearson Education, 2016.
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- **Overview**

Section 1 describes the purpose and scope of the system.

Section 2 explains overall description and user characteristics. Section 3

and 4 describe system interfaces and features.

- **Overall Description**

- **Product Perspective**

The application provides a graphical user interface for users to browse music easily. Screens

provided include:

- Login and Registration
- Home with music recommendations
- Search and Playlist screens
- **Product Functions**

Functions available to Users:

Account Maintenance – Allows users to create, update, and manage their personal profiles and login credentials securely.

Music Streaming – Enables users to listen to songs online in real time without downloading them.

Playlist Management – Allows users to create, edit, delete, and organize playlists according to their preferences.

Song Search – Provides users the ability to search songs by title, artist, album, or genre.

Offline Downloads (Premium) – Allows premium users to download songs and listen to them without an internet connection.

Like and Share Music – Enables users to like songs and share music or playlists with other users or on social platforms.

Functions available to Artists

- **Upload Songs –** Allows artists to upload their music files along with relevant details such as title, genre, and album information.
- **Manage Albums –** Enables artists to create, update, and organize albums by adding or removing songs.

Functions available to Admin

- **Manage Users** – Allows the administrator to monitor, add, modify, or remove user and artist accounts on the platform.
- **Manage Content and Subscriptions –** Enables the administrator to control music content availability, manage premium subscriptions, and monitor subscription plans.

- **User Characteristics**

User Classes:

Listeners – Access music and playlists

Artists – Upload and manage music **Admin** –

Maintain system

User Characteristics:

Users should understand basic English and be familiar with web or mobile applications.

- **Operating Environment**

The application works on Windows, Linux, Android, and iOS. Supported

browsers include Chrome, Firefox, Edge, and Safari. Internet connection

is required.

- **Design and Implementation Constraints**

• Internet connection required

• Secure login authentication

• Copyright and licensing constraints

• Large storage requirements

- **User Documentation**

The Online Music Application will provide help guides, and FAQs to assist users in understanding and using the system effectively. The documentation will include step-by-

step instructions for account management, music streaming, playlist creation, and subscription features. Support information will also be provided for troubleshooting common issues.

- **Assumptions and Dependencies**

• Users have internet access

• Artists upload legal content

• System depends on cloud storage

- **External Interface Requirements**
- **User Interfaces**

Login Screen

The Login Screen allows users to securely access the application by entering their registered credentials. It ensures that only authorized users can use personalized features such as playlists and recommendations.

The screen contains input fields for Username and Password. It also includes basic validation to prevent empty or incorrect entries.

For enhanced security, passwords are encrypted and authentication is handled through secure server communication.

Home Screen

The Home Screen is the main dashboard of the application. It provides users with a personalized experience immediately after login.

It displays Recommended Songs based on the user's listening history and preferences. This improves user engagement and satisfaction.

Additionally, the Recently Played section allows users to quickly access songs they have listened to before, enhancing convenience.

Playlist Screen

The Playlist Screen allows users to create and manage their personal music collections. Users can organize songs according to mood, genre, or preference.

Each playlist includes a Playlist Name and a Song List. Users can add, remove, or reorder songs within the playlist.

This feature improves user control over music organization and enhances the listening experience.

Subscription Screen

The Subscription Screen provides information about available premium plans. Users can view Plan Details such as pricing, features, and validity period.

It also displays the Payment Status to inform users whether their subscription is active, expired, or pending.

Secure payment gateways are integrated to ensure safe and reliable transactions.

- **Hardware Interfaces**

Desktop/Laptop

The application supports desktops and laptops through web browsers. This ensures users can access music from larger screens.

It is compatible with common operating systems such as Windows, macOS, and Linux.

Audio playback is supported through built-in or external speakers connected to the system.

Mobile Devices

The application is optimized for smartphones and tablets. It supports both Android and iOS platforms.

The mobile interface is responsive and touch-friendly to ensure smooth navigation. Users

can stream music on the go using mobile internet or Wi-Fi connectivity.

Audio Output Devices

The system supports various audio output devices such as headphones, earphones, Bluetooth speakers, and wired speakers.

It ensures high-quality sound output without distortion.

Compatibility with wireless devices enhances user convenience and flexibility.

- **Software Interfaces**

Web Browsers

The application is accessible through modern web browsers such as Chrome, Firefox, and Edge.

It uses standard web technologies like HTML, CSS, and JavaScript to ensure compatibility.

Browser-based access eliminates the need for installation on desktop devices.

Mobile Applications

Dedicated mobile applications provide enhanced performance and user experience. The app integrates with device features such as notifications and offline storage.

Regular updates ensure bug fixes and feature improvements.

Database Servers

The system connects to database servers to store user data, playlists, and music information.

It ensures efficient data retrieval for fast search and streaming.

Data security measures are implemented to protect user information.

- **Communication Interfaces**

Internet Connection

A stable internet connection is required for streaming music and accessing online features.

The application supports both Wi-Fi and mobile data networks.

Network optimization techniques are used to reduce buffering and improve streaming quality.

Secure HTTPS Communication

All communication between the client and server is secured using HTTPS protocol. This ensures encryption of user credentials and payment information.

Secure communication protects against unauthorized access and cyber threats.

- **System Features Music**

Streaming Feature

The Music Streaming Feature allows users to listen to songs online without downloading them.

It provides high-quality audio playback with minimal buffering using streaming technology.

The system adapts audio quality based on internet speed to ensure smooth performance.

Playlist Management Feature

This feature enables users to create, edit, and delete playlists.

Users can add or remove songs and organize them according to preference. Playlists

are saved in the user's account for easy access across devices.

Recommendation Feature

The Recommendation Feature suggests songs based on user listening history and preferences.

It uses algorithms to analyze patterns and generate personalized suggestions. This

improves user engagement and enhances the overall experience.

Offline Mode Feature

The Offline Mode Feature allows premium users to download songs for offline listening.

Downloaded content is securely stored within the application.

This feature is useful when internet connectivity is limited or unavailable.

- **Other Non-Functional Requirements**

- **Performance Requirements**

The Online Music Application must operate continuously and be available to users at all times without service interruption. Music playback should begin quickly after user selection to ensure a smooth listening experience. The system should efficiently handle multiple users accessing and streaming music simultaneously. Performance should remain stable even during peak usage hours. The application must ensure minimal buffering and latency.

- **Security Requirements**

The system must provide secure authentication to ensure that only authorized users can access their accounts. User data, including personal details and payment information, must be protected using encryption techniques. Regular data backups should be maintained to prevent loss of information due to system failures or cyberattacks. Security mechanisms should be in place to detect and prevent unauthorized access. The system should comply with standard security practices.

- **Safety Requirements**

The application will restrict system access to only valid and authenticated users through secure login credentials. Unauthorized attempts to access the system will be blocked to protect user accounts and data. Safety measures will ensure that user information is not altered or misused. Logging mechanisms will be maintained to track system usage and activities. These measures help safeguard the system from accidental or malicious misuse.

- **Software Quality Attributes**

The Online Music Application must be user-friendly so that users can easily navigate and use the system. The system should be reliable and provide uninterrupted music streaming with accurate content delivery. It should be portable across different platforms and devices

without loss of functionality. The application must be easy to maintain and support future updates. Additionally, the system should be scalable to handle an increasing number of users.

- **Business Rules**

All users must accept the terms and conditions before accessing the application services. Premium features such as offline downloads will be available only after successful subscription payment. The platform must strictly follow copyright and licensing regulations for all music content. Users are expected to use the application responsibly without violating policies. Any misuse of the platform may result in account suspension or termination.

- **Other Requirements**

The Online Music Application should support regular updates to improve features and fix issues. Customer support must be available to assist users with technical or account-related problems. A feedback mechanism should be provided to allow users to share suggestions and report issues. The system should continuously evolve based on user feedback. These requirements help improve overall system quality and user satisfaction.